

Abstract algebra – 4

Group actions, homomorphisms and automorphisms

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Theoretical minimum. A group is best understood through what it acts on. Two counting formulas — orbit–stabiliser and Burnside — and one construction — the action on cosets — cover almost all olympiad problems of this kind.

Definitions. An *action* of G on a set X is a homomorphism $G \rightarrow \text{Sym}(X)$, written $g \cdot x$. The *orbit* of x is $Gx = \{gx\}$, the *stabiliser* is $G_x = \{g : gx = x\}$, a subgroup. The action is *transitive* if there is one orbit and *faithful* if the homomorphism is injective. A homomorphism $\varphi: G \rightarrow H$ has kernel $\ker \varphi \trianglelefteq G$ and image $\varphi(G) \leq H$, and $G/\ker \varphi \cong \varphi(G)$. Automorphisms form the group $\text{Aut}(G)$.

Theorem 1 (orbit–stabiliser). $|Gx| = [G : G_x]$, because $gx = hx$ if and only if $g^{-1}h \in G_x$, so the map $gG_x \mapsto gx$ is a bijection from the cosets onto the orbit. In particular every orbit size divides $|G|$.

Theorem 2 (Burnside, or the Cauchy–Frobenius lemma). The number of orbits equals $\frac{1}{|G|} \sum_{g \in G} |\text{Fix}(g)|$. *Proof:* count the pairs (g, x) with $gx = x$ in two ways: by g it is $\sum_g |\text{Fix}(g)|$, by x it is $\sum_x |G_x| = \sum_{\text{orbits}} |Gx| \cdot \frac{|G|}{|Gx|} = |G| \cdot \#\text{orbits}$. The same identity will reappear in Abstract algebra – 5 as the statement that the averaging operator of the permutation representation has trace equal to the number of orbits.

Theorem 3 (Cayley and the action on cosets). G acts on itself by left translation; the action is faithful, so every group of order n embeds into S_n . More useful: G acts on the set G/H of cosets of a subgroup H of index n , giving a homomorphism $G \rightarrow S_n$ whose kernel is the *core* $\bigcap_g gHg^{-1}$, the largest normal subgroup of G contained in H . *Corollaries:* a subgroup whose index is the smallest prime divisor of $|G|$ is normal; a group with a subgroup of index n has a normal subgroup of index dividing $n!$.

Theorem 4 (conjugation). Conjugation is an action of G on itself; its orbits are the conjugacy classes, its stabilisers the centralisers, and the kernel is the centre. Hence $\text{Inn}(G) \cong G/Z(G)$, and $\text{Inn}(G) \trianglelefteq \text{Aut}(G)$.

Fact 1 (the automorphism groups one must know). $\text{Aut}(\mathbb{Z}/n) \cong (\mathbb{Z}/n)^*$ of order $\varphi(n)$; $\text{Aut}((\mathbb{Z}/p)^k) \cong \text{GL}_k(\mathbb{F}_p)$, of order $(p^k - 1)(p^k - p) \cdots (p^k - p^{k-1})$. The size of Aut is not monotone in the size of the group, which is the point of problem 3.

1. Prove the orbit-stabiliser theorem and Burnside's lemma. Using them, count the necklaces of n beads in k colours up to rotation, and deduce that $\frac{1}{n} \sum_{d|n} \varphi(d) k^{n/d}$ is an integer for all n, k .
2. Prove Cayley's theorem and the coset-action theorem, and deduce that a subgroup whose index is the smallest prime divisor of $|G|$ is normal. Then prove, by looking at the sign of the permutation induced by left translation, that a group of order $2m$ with m odd has a subgroup of index 2.
3. Does there exist a finite group G with a normal subgroup H such that $|\text{Aut } H| > |\text{Aut } G|$? (IMC)
4. Let G be a group with identity e and $\varphi: G \rightarrow G$ a map such that $\varphi(g_1)\varphi(g_2)\varphi(g_3) = \varphi(h_1)\varphi(h_2)\varphi(h_3)$ whenever $g_1g_2g_3 = e = h_1h_2h_3$. Prove that there is $a \in G$ for which $\psi(x) = a\varphi(x)$ is a homomorphism. (Putnam 1997, A4)
5. Let G be a group, $n \geq 2$, and let $H_1, H_2 \leq G$ satisfy $[G : H_1] = [G : H_2] = n$ and $[G : (H_1 \cap H_2)] = n(n-1)$. Prove that H_1 and H_2 are conjugate in G . (IMC)